

Q5a) Explain Different Types of Arrays in PHP with Example

Array in PHP - An array is a data structure used to store multiple values in a single variable.

Types of Arrays in PHP - There are mainly three types of arrays in PHP: Indexed Array, Associative Array, Multidimensional Array

1) Indexed Array - Indexed arrays store elements with numeric indexes starting from 0.

Example - <?php
\$colors = array("Red","Green","Blue");
echo \$colors[0];
echo "
";
echo \$colors[1];
?>

Output: Red
Green

2) Associative Array - Associative arrays store data in key-value pairs.

Example - <?php
\$student = array(
"name"=>"Rahul",
"roll"=>101,
"dept"=>"Computer"
);
echo \$student["name"];
echo "
";
echo \$student["dept"];
?>

Output: Rahul
Computer

3) Multidimensional Array - Multidimensional arrays contain one or more arrays inside another array.

Example - <?php
\$emp = array(
array(101,"Rahul","IT"),
array(102,"Amit","HR"),
array(103,"Neha","Sales")
);
echo \$emp[0][1];

```
echo "<br>";  
echo $emp[1][2];  
?>
```

Output: Rahul

HR

Advantages of Arrays - Stores multiple values, Easy data management, Reduces coding complexity, Supports dynamic data handling

PHP supports indexed, associative, and multidimensional arrays for efficient storage and management of multiple data values.

Q6a) Write PHP Code to Connect MySQL Database to Display Records from the Table

PHP MySQL Connectivity - PHP uses MySQL functions to connect and interact with MySQL database.

Steps for PHP-MySQL Connectivity -

1. Connect database
2. Execute SQL query
3. Fetch records
4. Display output
5. Close connection

PHP Program -

```
<?php  
// Database Connection  
$conn = mysqli_connect(  
"localhost",  
"root",  
"",  
"test"  
);  
  
// Check Connection  
if(!$conn)  
{  
    die("Connection Failed");  
}  
  
// SQL Query
```

```

$sql = "SELECT * FROM employee";

// Execute Query
$result = mysqli_query($conn,$sql);

echo "<table border='1'>";

echo "<tr>";
echo "<th>ID</th>";
echo "<th>Name</th>";
echo "<th>Department</th>";
echo "</tr>";

// Fetch Records
while($row = mysqli_fetch_assoc($result))
{
    echo "<tr>";
    echo "<td>".$row['emp_id']. "</td>";
    echo "<td>".$row['emp_name']. "</td>";
    echo "<td>".$row['emp_dept']. "</td>";
    echo "</tr>";
}

echo "</table>";

// Close Connection
mysqli_close($conn);
?>

```

Output -

ID	Name	Department
101	Rahul	IT
102	Amit	HR
103	Neha	Sales

Advantages - Dynamic database interaction, Easy record management, Supports web applications

PHP can easily connect with MySQL database using MySQLi functions to retrieve and display records dynamically.

Q6b) Draw and Explain .NET Framework

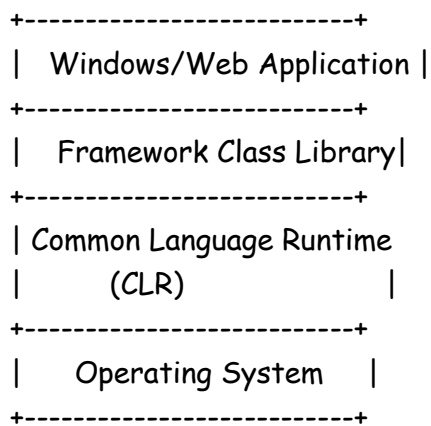
.NET Framework is a software development platform developed by Microsoft for building and running applications.

It supports multiple programming languages such as: C#, VB.NET, F#

Components of .NET Framework -

1. Common Language Runtime (CLR)
2. Framework Class Library (FCL)

.NET Framework Architecture Diagram



1) Common Language Runtime (CLR) - CLR is the execution engine of .NET framework. It provides: Memory management, Exception handling, Security, Garbage collection.

Functions of CLR - Executes programs, Manages memory, Handles exceptions, Converts code into machine code

2) Framework Class Library (FCL) - FCL is a collection of reusable classes and libraries. It provides classes for: File handling, Database connectivity, GUI development, Networking. **Features of .NET Framework** - Platform support, Language interoperability, Automatic memory management, Security support, Rich class libraries

Advantages of .NET Framework - Fast development, Reusable code, Easy debugging, Better security, Supports web and desktop applications

Applications of .NET - Web applications, Windows applications, Enterprise software, Cloud applications

.NET Framework provides a secure and efficient environment for developing and executing applications using CLR and extensive class libraries.

Q5a) Explain the MySQL & PHP for Inserting the Students Data into the Students Table

PHP and MySQL Connectivity - PHP can interact with MySQL database using MySQLi functions. It is used to: Insert records, Retrieve records, Update records, Delete records

Step 1: Create Students Table

```
CREATE TABLE students (  
    roll_no INT,  
    name VARCHAR(20),  
    department VARCHAR(20)  
);
```

Step 2: HTML Form for Student Data

```
<html>  
<body>  
<form method="post">  
  
Roll No :  
<input type="text" name="roll"><br><br>  
  
Name :  
<input type="text" name="name"><br><br>  
  
Department :  
<input type="text" name="dept"><br><br>  
  
<input type="submit" value="Insert">  
  
</form>  
</body>  
</html>
```

Step 3: PHP Code for Database Insertion

```
<?php  
// Database Connection  
$conn = mysqli_connect(  
    "localhost",  
    "root",  
    "",  
    "college"  
);  
  
// Check Connection
```

```

if(!$conn) {
    die("Connection Failed");
}

// Get Form Data
$roll = $_POST['roll'];
$name = $_POST['name'];
$dept = $_POST['dept'];

// SQL Insert Query
$sql = "INSERT INTO students
VALUES('$roll','$name','$dept')";

// Execute Query
if(mysqli_query($conn,$sql)) {
    echo "Record Inserted Successfully";
} else {
    echo "Insertion Failed";
}

// Close Connection
mysqli_close($conn);
?>

```

Working -

1. User enters student details in form
2. PHP receives data using \$_POST
3. PHP connects with MySQL database
4. SQL INSERT query executes
5. Student data is stored in table

Output - Record Inserted Successfully

Advantages - Dynamic data storage, Easy database management, Supports web applications

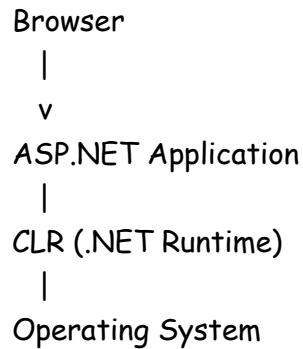
PHP and MySQL together provide an efficient way to store and manage student information dynamically in database tables.

Q5b) Write Note on: i) ASP.NET ii) NodeJS

i) **ASP.NET** is a web application framework developed by Microsoft for building dynamic web applications and web services. It is a part of the .NET framework.

Features of ASP.NET - Server-side technology, Supports dynamic web pages, Object-oriented programming, Built-in security, Easy database connectivity

ASP.NET Architecture



Components of ASP.NET - CLR (Common Language Runtime), ASP.NET Libraries, Web Forms, MVC Support

Advantages of ASP.NET - Fast development, Better security, Easy deployment, Rich controls and libraries

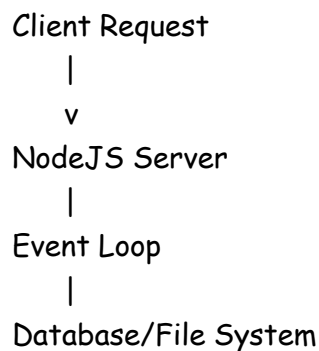
Applications - E-commerce websites, Enterprise applications, Banking systems

ASP.NET provides a secure and efficient environment for developing dynamic and enterprise-level web applications.

ii) **NodeJS** is an open-source JavaScript runtime environment used for developing server-side applications. It executes JavaScript code outside the browser.

Features of NodeJS - Event-driven, Non-blocking I/O, Fast execution, Single-threaded, Scalable applications

Architecture of NodeJS



Example of NodeJS -

```
const http = require('http');
http.createServer((req,res)=> {
  res.write("Welcome to NodeJS");
  res.end();
}).listen(3000);
```

Advantages of NodeJS - High performance, Lightweight, Fast data processing, Suitable for real-time applications

Applications of NodeJS - Chat applications, Streaming applications, REST APIs, Real-time systems

NodeJS is widely used for developing scalable and high-performance server-side web applications using JavaScript.

Q6a) Explain PHP Functions and Arrays in PHP

PHP Functions - A function in PHP is a block of reusable code used to perform a specific task.

Functions help: Reduce code duplication, Improve readability, Simplify program maintenance

Types of Functions in PHP

1. Built-in Functions
2. User-defined Functions

1) Built-in Functions - PHP provides many predefined functions.

Examples: strlen(), sqrt(), count(), date()

Example - <?php
 \$str = "Welcome";
 echo strlen(\$str);
 ?>

Output - 7

2) User-defined Functions - Functions created by programmer are called user-defined functions.

Syntax -

```
function function_name() {
  // code
}
```


Example - <?php
function add(\$a, \$b) {
 \$sum = \$a + \$b;
 echo "Sum = ".\$sum;
}
add(10,20);
?>

Output - Sum = 30

Advantages of Functions - Code reusability, Easy debugging, Better modularity, Reduces complexity

Q6b) Explain the Concept of .NET Framework and C#

C# (C Sharp) is an object-oriented programming language developed by Microsoft.

It is mainly used with .NET framework for developing: Web applications, Desktop applications, Enterprise software

Features of C# - Object-oriented, Simple syntax, Type safe, Automatic garbage collection, Platform independent with .NET

Example of C# Program -

```
using System;  
class Program {  
    static void Main() {  
        Console.WriteLine("Hello C#");  
    }  
}
```

Output - Hello C#

Advantages of C# - Easy to learn, Secure programming , Fast application development, Rich library support

Applications of C# - Web applications, Windows applications, Game development, Enterprise software

.NET Framework provides a platform for application development, while C# is a powerful object-oriented programming language used for building secure and scalable applications.

Q5b) Explain the Following

i) WAP & WML

WAP stands for: Wireless Application Protocol

WAP is a protocol used for accessing internet services on mobile and wireless devices.

It enables mobile phones to browse web pages and access online services.

Features of WAP - Supports wireless communication, Optimized for mobile devices, Lightweight protocol, Provides internet access on mobile phones

WAP Architecture

Mobile Device

|

v

WAP Gateway

|

v

Web Server

Components of WAP - WAP Browser, WAP Gateway, WAP Server, WML

Advantages of WAP - Mobile internet access, Lightweight communication, Faster data transfer on wireless networks

WML stands for: Wireless Markup Language

WML is a markup language used for creating web pages for WAP-enabled mobile devices.

It is similar to HTML but designed for small-screen devices.

Features of WML - XML based language, Lightweight pages, Supports mobile devices, Small memory requirement

Example of WML -

```
<?xml version="1.0"?>
<wml>
<card id="c1" title="Welcome">
<p>Hello User</p>
</card>
</wml>
```

Advantages of WML - Simple syntax , Optimized for mobile phones, Low bandwidth usage

WAP provides wireless internet communication, while WML is used to create web pages for mobile devices using WAP technology.

ii) Basis	C#	Java
Developed By	Microsoft	Oracle
Platform	Mainly .NET platform	JVM based platform
File Extension	.cs	.java
Compilation	Compiles into MSIL	Compiles into bytecode
Runtime Environment	CLR	JVM
Pointer Support	Supports pointers	No direct pointer support
Property Support	Available	Not directly available
Operator Overloading Supported		Limited support
Multiple Inheritance	Through interfaces	Through interfaces
GUI Framework	Windows Forms, WPF	Swing, JavaFX

Both C# and Java are object-oriented programming languages, but C# is closely associated with .NET framework while Java focuses on platform independence using JVM.

Q6b) Write Note on Session Tracking in PHP

Session Tracking in PHP - Session tracking is used to maintain user information across multiple web pages.

HTTP is stateless, so session tracking helps remember user data during interaction.

Need of Session Tracking - User login management, Shopping cart, Online banking, Personalized websites

PHP Session Functions

Function	Purpose
session_start()	Starts session
\$_SESSION[]	Stores session data
session_destroy()	Ends session

Example Program:

Page 1 -

```
<?php
session_start();
$_SESSION["user"] = "Mayur";
echo "Session Created";
?>
```

Page 2 -

```
<?php
session_start();
echo $_SESSION["user"];
?>
```

Output - Mayur

Working of PHP Session

1. User accesses webpage
2. Server creates unique session ID
3. Session data stored on server
4. Session ID stored in browser
5. User information maintained across pages

Session Tracking Diagram

```
Browser
|
Session ID
|
Web Server
|
Session Data
```

Advantages of Session Tracking - Maintains user state, More secure than cookies, Easy user management

Disadvantages - Uses server memory, Session timeout problems

Session tracking in PHP helps maintain user-specific information across multiple pages and is widely used in dynamic web applications.

Q6a) Explain How Cookies and Sessions are Used for Session Management in PHP

Session Management in PHP - HTTP protocol is stateless, meaning the server does not remember previous requests.

Therefore, PHP uses: Cookies & Sessions to maintain user information across multiple web pages.

1) Cookies in PHP - Cookies are small text files stored on the client browser.

They store user-related information such as: Username, Preferences, Login information

Working of Cookies

1. Server creates cookie
2. Cookie stored in browser
3. Browser sends cookie with future requests
4. Server identifies user using cookie data

Syntax of Cookie - `setcookie(name,value,time);`

Example of Cookie -

```
<?php
setcookie("user", "Mayur", time() + 3600);
echo "Cookie Created";
?>
```

Reading Cookie -

```
<?php
echo $_COOKIE["user"];
?>
```

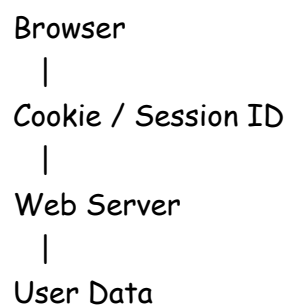
Advantages of Cookies - Easy to implement, Stores small user data, Automatic browser storage

Disadvantages of Cookies - Less secure, User can disable cookies, Limited storage size

2) Sessions in PHP ...

Cookies	Sessions
Stored in browser	Stored on server
Less secure	More secure
Small storage	Larger storage
Can be disabled	Cannot be easily disabled

Session Management Diagram -



Cookies and sessions are widely used in PHP for maintaining user information and providing continuous interaction across multiple web pages.

Q5a) Explain the Following with Respect to PHP

i) **Control Statements in PHP** - Control statements in PHP are used to control the flow of program execution. They help in: Decision making, Repetition of statements, Program branching

Types of Control Statements in PHP - Conditional Statements, Looping Statements, Jump Statements.

1) **Conditional Statements** - Used for decision making.

a) **if Statement** - Executes block if condition is true.

Syntax -	Example - <?php
	\$a = 10;
if(condition) {	if(\$a > 5) {
statements;	echo "A is Greater";
}	}
	?>

Output - A is Greater

b) **if-else Statement** - Executes one block if condition is true otherwise another block.

Example - <?php
\$a = 5;
if(\$a > 10) {
 echo "Greater";
} else {
 echo "Smaller";
}
?>

Output - Smaller

c) **switch Statement** - Used for multiple conditions.

Example - <?php
\$day = 2;
switch(\$day) {
 case 1:
 echo "Monday";
 break;
 case 2:
 echo "Tuesday";
 break;
 default:
 echo "Invalid";
}

```
}  
?>
```

Output - Tuesday

2) Looping Statements -

Used to execute statements repeatedly.

a) for Loop - <?php
for(\$i=1;\$i<=5;\$i++){
 echo \$i."
";
}
?>

b) while Loop

```
<?php  
$i=1;  
while($i<=5) {  
    echo $i."<br>";  
  
    $i++;  
}  
?>
```

c) foreach Loop - Used mainly with arrays.

```
<?php  
$colors = array("Red","Blue");  
foreach($colors as $c) {  
    echo $c."<br>";  
}  
?>
```

3) Jump Statements - Used to transfer control.

Examples: break, continue, exit

Advantages of Control Statements - Simplifies programming, Supports decision making , Reduces code repetition

Control statements in PHP help manage program execution efficiently using conditions, loops, and branching mechanisms.